

# SONAR-X

Physics-Aware Explainable Marine Debris Intelligence System

## SURVEY INTELLIGENCE REPORT

### Survey Information

|             |                               |
|-------------|-------------------------------|
| Survey ID   | SX-SRV-20260924-005           |
| Survey Name | SONAR Survey — e2e sonar test |
| Survey Date | 2026-09-24                    |
| Area        | Not specified                 |
| Operator    | Not specified                 |
| Data Source | SURVEY                        |
| Status      | COMPLETED                     |
| Generated   | 2026-09-24 17:39 UTC          |
| Software    | SONAR-X 0.1.0-prototype       |

### Sonar Metadata

|                               |               |
|-------------------------------|---------------|
| Altitude above seabed         | Not available |
| Range scale                   | Not available |
| Channel layout                | BOTH          |
| Already gain corrected        | No            |
| Already slant-range corrected | No            |

### Detection Summary

|                       |   |
|-----------------------|---|
| Total detections      | 5 |
| Marine Debris         | 0 |
| Natural Formation     | 0 |
| Sonar Artifact        | 0 |
| Unknown Anomaly       | 5 |
| Hazard — High Caution | 4 |
| Hazard — Caution      | 1 |

### Detection List

| Detection UID                  | Class           | Conf.* | Evidence* | Risk   | Hazard  | Lat | Lon | Status  |
|--------------------------------|-----------------|--------|-----------|--------|---------|-----|-----|---------|
| SX-DET-SX-SRV-20260924-005-005 | UNKNOWN ANOMALY | 47     | 47        | MEDIUM | CAUTION | N/A | N/A | PENDING |

| Detection UID                  | Class           | Conf.* | Evidence* | Risk   | Hazard       | Lat | Lon | Status  |
|--------------------------------|-----------------|--------|-----------|--------|--------------|-----|-----|---------|
| SX-DET-SX-SRV-20260924-005-004 | UNKNOWN ANOMALY | 69     | 69        | MEDIUM | HIGH_CAUTION | N/A | N/A | PENDING |
| SX-DET-SX-SRV-20260924-005-003 | UNKNOWN ANOMALY | 83     | 83        | MEDIUM | HIGH_CAUTION | N/A | N/A | PENDING |
| SX-DET-SX-SRV-20260924-005-002 | UNKNOWN ANOMALY | 64     | 64        | MEDIUM | HIGH_CAUTION | N/A | N/A | PENDING |
| SX-DET-SX-SRV-20260924-005-001 | UNKNOWN ANOMALY | 66     | 66        | MEDIUM | HIGH_CAUTION | N/A | N/A | PENDING |

\* Confidence and Evidence Score are prototype scores, NOT validated probabilities.

## ■ HAZARD ASSESSMENT

The following detections triggered precautionary hazard flags. This system does NOT identify specific hazardous objects. "Potential Hazard — Do Not Disturb" means sonar evidence warrants human review. Absence of a hazard flag does NOT mean an object is safe.

### ● SX-DET-SX-SRV-20260924-005-005 — CAUTION — UNKNOWN\_ANOMALY

Indicators: Regular / structured geometry, Limited metadata / reduced signal quality

### ● SX-DET-SX-SRV-20260924-005-004 — HIGH\_CAUTION — UNKNOWN\_ANOMALY

Indicators: Regular / structured geometry, Strong acoustic backscatter, Unknown class with strong acoustic evidence, Limited metadata / reduced signal quality

### ● SX-DET-SX-SRV-20260924-005-003 — HIGH\_CAUTION — UNKNOWN\_ANOMALY

Indicators: Regular / structured geometry, Strong acoustic backscatter, Unknown class with strong acoustic evidence, Limited metadata / reduced signal quality

### ● SX-DET-SX-SRV-20260924-005-002 — HIGH\_CAUTION — UNKNOWN\_ANOMALY

Indicators: Regular / structured geometry, Strong acoustic backscatter, Unknown class with strong acoustic evidence, Limited metadata / reduced signal quality

### ● SX-DET-SX-SRV-20260924-005-001 — HIGH\_CAUTION — UNKNOWN\_ANOMALY

Indicators: Large spatial extent, Regular / structured geometry, Strong acoustic backscatter, Unknown class with strong acoustic evidence, Limited metadata / reduced signal quality

## Limitations and Disclaimers

- This is a prototype system. All detections require human verification before any action.
- Confidence and evidence scores are prototype heuristic values, NOT validated probabilities.
- Side-scan sonar does not directly provide water temperature or depth.
- Conventional side-scan sonar is not a direct bathymetric sensor.
- Slant-range correction geometry is approximate where metadata was unavailable.
- Geolocation is approximate. GPS coordinates require field calibration.
- A missing detection does NOT prove debris was removed or moved.
- Absence of a hazard flag does NOT mean an object is safe.

**AI-generated observations require verification by qualified human experts.**